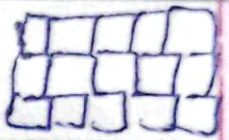
Convulational N.N :- 1998.

Class of N.N used for the image based data.

→ Grid like time series data  

→ In ANN → matrix multiplication

" CNN → Convulation operation

Inspired
└ Visual cortex.

CNN >> ANN on image data.

CamScanner

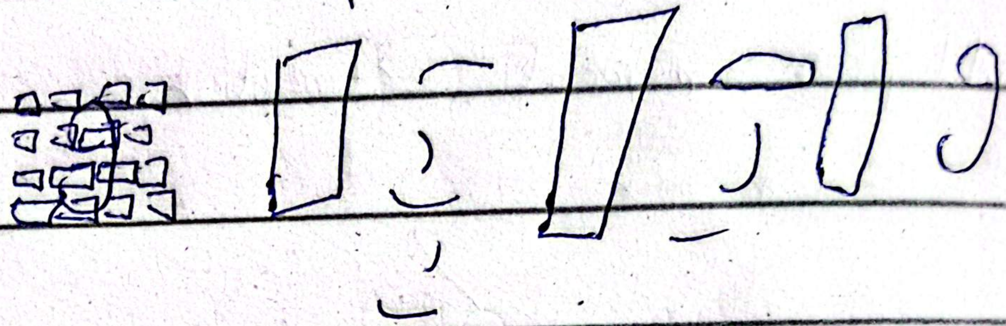
Convulations layer → filters

features extract.

Day:

Date:

→ It catches patterns in data



Roadmap:-

→ Biological connection

→ Convulation operation, padding, stride.

→ pooling.

→ Dog vs cat classifier.

→ Data argument.

→ popular CNN architecture.

→ CNN work best on images as it is based on our own visual cortex.

→ Image is a collection of pixels.

1 channel in black & white.
3 " " Red, green, blue. (RGB)

Date: Day:

Convulation operation:-

0 0 0	-1 -1 -1	☐ ☐ ☐
0 0 0	0 0 0	☐ ☐ ☐
255 255 255	1 1 1	
image.	filter	feature map.

6×6	3×3	$\boxed{4\times 4}$
n×n	f×f	(n-f+1) (n-f+1)
5×5	3×3	(5-3+1) (3×3)

Padding:-

We want $n-f+1 = 5$ in padding.

→ for this we add row & coloumns above & below & on both sides & that pixel values is 0 called zero padding.

So formule now $(n+2p-f+1)$

Strides:-

moving 1 pixel to other is called stride.

U can set it (1,1) or (2,2) etc.
→ Strided convulation

Date: Day:

Pooling:-

issues in con.net:-

1) Memory issue — u need to reduce size

2) Translation varionce.

A way to down sample your feature map is called pooling.

-> Through this features b/m location independ.

3	1	1	3
2	5	0	2
1	4	2	1
4	7	2	4

5	3
7	4

max pooling.

Capturing dominent features

-> Translation invariance achieved.

-> Another advant is no need of training.

Disadvan:- lose a lot of information means in max pooling around 75%.

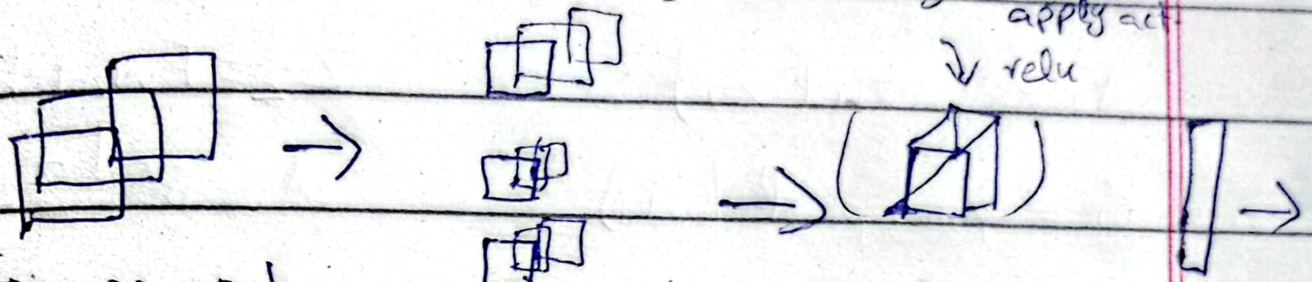
-> Not good in image segmentation.

CS CamScanner

Date: ____ Day: ____

CNN Architecture:- [Lenet-5]:-

apply act
relu

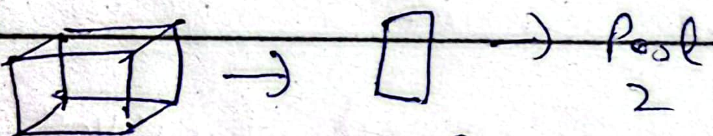


(32×32×3)
input (RGB image)

con-layer.
filters or kernels.
con-1.

feat map.

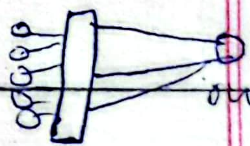
Pooling layer.
Pool 1



con2.

Pool 2 → flatten tensor.

1D



output lay.

Fully con. layer

Logical flow:- (B.P)

$$X \to \underset{W_1, b_1}{\otimes} \to Z_1 \xrightarrow{relu} A_1 \xrightarrow[pool]{max} P_1 \xrightarrow{flatten} F \to$$

$$Z_2 \xrightarrow{sigmoid} A_2 \to L$$

Forward Prop:-

relu diff | + 1
− 0.

Date: Day:

$Z_1 = \text{Conv}(x, w_1) + b_1$ F (flatten P_1)

$A_1 = \text{relu}(z_1)$ $Z_2 = F w_2 + b_2$

$P_1 = \text{maxpool}(A_1)$ $A_2 = \sigma(z_2)$

→ We have to apply Grad. descent & our objective is to find values to minimize loss for (w_1, w_2) (b_1, b_2

B.P :- (CNN) :-

Divide it into 2 categories of CNN & ANN.

So 1st find of ANN.

$$\frac{\partial L}{\partial W_2} = (A_2 - Y) F^T, \quad \frac{\partial L}{\partial b_2} = A_2 - Y.$$

$$\frac{\partial L}{\partial W_1} = \text{Conv}\left(\underset{\text{input}}{X}, \underset{\text{loss}}{\frac{\partial L}{\partial Z_1}}\right)$$

$$\frac{\partial L}{\partial b_1} = \text{Sum}\left(\frac{\partial L}{\partial Z_1}\right)$$

& that's how B.P performs on CNN

ImageDataGenerator-

mobilenet

Date:

Day:

Data argumentation:-

Creating new data from the available less amount of data.

-> Used to generate data.

-> To reduce overfitting.

Pretrained models:-

Already trained model.

Why we use:-

-> D.L model data hungry.

-> Time -> model -> training.

Already made model like VGGNet ResNet we can use -> Have trained parameters.

CS CamScanner